

Synthetic Data Generation - Technical Implementation Guide

Introduction

This document outlines the technical implementation of synthetic data generation workflows for machine learning practitioners. The goal is to enable scalable, reproducible, and privacy-compliant synthetic data pipelines.

Pipeline Design

A robust synthetic data pipeline typically includes the following stages: 1. Data Profiling: Analyze distributions, correlations, and feature importance. 2. Model Selection: Choose between GANs, VAEs, or tabular synthesis models like CTGAN. 3. Generation: Train the model using anonymized or sample data. 4. Validation: Compare real vs synthetic distributions using statistical metrics such as KS-test, Chi-square, or Wasserstein distance. 5. Deployment: Serve synthetic data through APIs or data catalogs.

Tools and Frameworks

Several open-source tools support synthetic data generation: - SDV (Synthetic Data Vault): Supports relational, sequential, and tabular data synthesis. - Gretel.ai: Provides privacy-focused synthetic data generation as a service. - ydata-synthetic: Offers GAN-based tabular synthesis capabilities. - Synthcity: Framework for benchmarking and comparing multiple synthesis models.

Example Use Case

Consider a financial institution that cannot share customer transaction data. By training a CTGAN model on a sanitized subset, they can produce realistic synthetic transaction records. These records can be used to train fraud detection models without compromising user privacy.

Evaluation Metrics

To ensure synthetic data quality: - Fidelity: Evaluate how well synthetic data preserves real data structure. - Utility: Measure downstream ML performance on synthetic vs real data. - Privacy: Assess risk of data leakage via membership inference or re-identification attacks.